

5 (long) A large packaging machine consumes 200 kW of power when a 1 kV, 50 Hz supply is directly connected across it. The machine has a 0.85 lagging power factor. A feeder line with a complex impedance of $(0.8+j1.6) \Omega$ now supplies the machine from a 50 Hz voltage source.

(a) If the voltage at the machine terminals is 1 kV, calculate the feeder current, feeder power loss, and the source voltage. [12]

(b) Power factor correction is applied by connecting a capacitor in parallel with the machine. If 1 kV is maintained across the machine terminals, calculate the required value of this capacitor, the new feeder current, and the feeder power loss. [10]

(c) With the power factor correction capacitor still connected, if the source voltage is set at 1 kV, what will be the magnitude of the voltage across the machine terminals? [8]